

# Simultaneous Differential Equations

Example:

$$x'(t) + 2y'(t) + 2x(t) + 5y(t) = 77$$

$$y'(t) + x(t) + 4y(t) = 61$$

Solution:-

$$x'(t) + 2y'(t) + 2x(t) + 5y(t) = 77$$

$$y'(t) + x(t) + 4y(t) = 61$$

Write in matrix notation as:

$$Ju + Mv = g$$

where the matrices are

$$J = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$u = \begin{bmatrix} x'(t) \\ y'(t) \end{bmatrix}$$

$$M = \begin{bmatrix} 2 & 5 \\ 1 & 4 \end{bmatrix}$$

$$v = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$$

$$g = \begin{bmatrix} 77 \\ 61 \end{bmatrix}$$



$$= \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x'(t) \\ y'(t) \end{bmatrix} + \begin{bmatrix} 2 & 5 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} 77 \\ 61 \end{bmatrix}$$

For  $\lambda_1, \lambda_2$  use the formula:

$$|\lambda J + M| = 0$$

$$\left| \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 5 \\ 1 & 4 \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} \lambda + 1 & 2\lambda \\ 0 & \lambda + 1 \end{bmatrix} + \begin{bmatrix} 2 & 5 \\ 1 & 4 \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} \lambda + 2 & 2\lambda + 5 \\ 0 + 1 & \lambda + 4 \end{bmatrix} \right| = 0$$

$$\left| \begin{array}{cc|c} \lambda + 2 & 2\lambda + 5 & \\ \hline 1 & \lambda + 4 & \end{array} \right| = 0$$

$$(\lambda + 2)(\lambda + 4) - (1 \times 5) = 0$$

$$\lambda^2 + 4\lambda + 2\lambda + 8 - 5 = 0$$

$$\lambda^2 + 6\lambda + 3 = 0$$

$$\lambda^2 + 4\lambda + 3 = 0$$



Factorization:

$$x^2 + 4x + 3 = 0$$

$$x^2 + 3x + x + 3 = 0$$

$$x(x+3) + 1(x+3) = 0$$

$$(x+1)(x+3) = 0$$

$$x+1 = 0 \Rightarrow \boxed{x_1 = -1}$$

$$x+3 = 0 \Rightarrow \boxed{x_2 = -3}$$

Complementary Function

For  $C_1, C_2$  we use eigenvalue formula:

$$(xI + M)C_i = 0$$

$$x \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 5 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} xI & 2x \\ 0 & xI \end{bmatrix} + \begin{bmatrix} 2 & 5 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = 0$$



$$\begin{bmatrix} x_1 + 2 & 2x + 5 \\ 0 + 1 & x_1 + 4 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} x_1 + 2 & 2x + 5 \\ 1 & x_1 + 4 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = 0 \text{ --- eq. (i)}$$

Put the value  $x_1 = -1$  in eq. (i)

$$\begin{bmatrix} -1 + 2 & 2(-1) + 5 \\ -1 + 1 & -1 + 4 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} 1 & 3 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = 0$$

$$1c_1 + 3c_2 = 0$$

$$c_1 + 3c_2 = 0$$

Now assume  $c_2 = 1$

$$c_1 + 3c_2 = 0$$

$$c_1 + 3(1) = 0$$

$$c_1 + 3 = 0$$

$$\boxed{c_1 = -3}$$



$$c_1 + 4c_2 = 0$$

$$-3 + 4c_2 = 0$$

$$\boxed{c_2 = 1}$$

Thus, the eigenvector  $c_1$  corresponding to  $\lambda_1 = -1$  is

$$c_1 = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

First complementary function

$$y e^t = k_1 \begin{bmatrix} -3 \\ 1 \end{bmatrix} e^{-t} = \begin{bmatrix} -3k_1 e^{-t} \\ k_1 e^{-t} \end{bmatrix}$$

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Now next step

Put the value of  $\lambda_2 = -3$  in eq. (i)

$$\begin{bmatrix} -3+2 & 2(-3)+5 \\ 1 & -3+4 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = 0$$



$$\begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = 0$$

$$-c_1 - c_2 = 0$$

$$c_1 + c_2 = 0$$

Now assume  $c_2 = 1$

$$c_1 + c_2 = 0$$

$$c_1 + 1 = 0$$

$$\boxed{c_1 = -1}$$

$$c_1 + c_2 = 0$$

$$-1 + c_2 = 0$$

$$\boxed{c_2 = 1}$$

Thus the eigenvector  $c_2$  corresponding to  $\lambda_2 = -3$  is

$$c_2 = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Second complementary function



$$V^2 c = k_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{-3t} = \begin{bmatrix} -k_2 e^{-3t} \\ k_2 e^{-3t} \end{bmatrix}$$

Putting these values for the complete complementary function:

$$y_1(t) = -3k_1 e^{-4t} - k_2 e^{-3t}$$

$$y_2(t) = k_1 e^{-4t} + k_2 e^{-3t}$$

Particular Integral

$$y_p = A^{-1} B$$

where,

$$B = \begin{bmatrix} 7 \\ 6 \end{bmatrix}, A = \begin{bmatrix} 2 & 5 \\ 1 & 4 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 2 & 5 \\ 1 & 4 \end{vmatrix}$$

$$|A| = (2 \times 4) - (1 \times 5)$$

$$|A| = 8 - 5$$

$$\boxed{|A| = 3}$$



$$A^{-1} = \frac{\text{Adj } A}{|A|}$$

$$\frac{\text{Adj } A}{|A|} = \frac{\begin{bmatrix} 4 & -5 \\ -1 & 2 \end{bmatrix}}{3}$$

$$A^{-1} = \frac{1}{3} \begin{bmatrix} 4 & -5 \\ -1 & 2 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 4/3 & -5/3 \\ -1/3 & 2/3 \end{bmatrix}$$

$$\begin{aligned} \forall p &= A^{-1}B \\ &= \begin{bmatrix} 4/3 & -5/3 \\ -1/3 & 2/3 \end{bmatrix} \begin{bmatrix} 77 \\ 61 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} (4 \times 77)/3 + (-5 \times 61)/3 \\ (-1 \times 77)/3 + (2 \times 61)/3 \end{bmatrix}$$

$$= \begin{bmatrix} 308/3 + -305/3 \\ -77/3 + 122/3 \end{bmatrix}$$



$$= \begin{bmatrix} 3/8 \\ 15/8 \end{bmatrix}$$

$$y_p = \begin{bmatrix} 1 \\ 15 \end{bmatrix}$$

Complete General Solution

$$y(t) = y_c + y_p$$

$$y_1(t) = -3k_1 e^{-4t} - k_2 e^{-3t} + 1$$

$$y_2(t) = 15k_1 e^{-4t} + k_2 e^{-3t} + 15 \quad \underline{\text{Ans}}$$

### Exercise 19.2

Solve the following two differential-equation system:

$$(a) \quad x'(t) - x(t) - 12y(t) = 60$$

$$y'(t) + x(t) + 6y(t) = 36$$

[with  $x(0) = 13$  and  $y(0) = 4$ ]



**Solution:**

$$x'(t) - x(t) - 12y(t) = -60$$

$$y'(t) + x(t) + 6y(t) = 36$$

Write in matrix equation

$$A_1 y' + A_2 y = B$$

where the matrices are:

$$A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, y' = \begin{bmatrix} x'(t) \\ y'(t) \end{bmatrix}, A_2 = \begin{bmatrix} -1 & -12 \\ 1 & 6 \end{bmatrix},$$

$$y = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}, B = \begin{bmatrix} -60 \\ 36 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x'(t) \\ y'(t) \end{bmatrix} + \begin{bmatrix} -1 & -12 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} -60 \\ 36 \end{bmatrix}$$

For  $x_1, x_2$  use the formula



$$|A_1 \lambda_i + A_2| = 0$$

$$\left| \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \lambda_i + \begin{bmatrix} -1 & -12 \\ 1 & 6 \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} \lambda_i & 0 \\ 0 & \lambda_i \end{bmatrix} + \begin{bmatrix} -1 & -12 \\ 1 & 6 \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} \lambda_i - 1 & 0 - 12 \\ 0 + 1 & \lambda_i + 6 \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} \lambda_i - 1 & -12 \\ 1 & \lambda_i + 6 \end{bmatrix} \right| = 0$$

$$(\lambda - 1)(\lambda + 6) - (1 \times -12) = 0$$

$$\lambda^2 + 6\lambda - \lambda - 6 + 12 = 0$$

$$\lambda^2 + 5\lambda + 6 = 0$$

Factorization:

$$\lambda^2 + 5\lambda + 6 = 0$$

$$\lambda^2 + 2\lambda + 3\lambda + 6 = 0$$



$$s(s+2) + 3(s+2) = 0$$

$$(s+3)(s+2) = 0$$

$$s+3 = 0 \Rightarrow \boxed{s_1 = -3}$$

$$s+2 = 0 \Rightarrow \boxed{s_2 = -2}$$

## Complementary Function

For  $C_1, C_2$  we use eigenvector formula:

$$(A_1 s^i + A_2) C_i = 0$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} s^i + \begin{bmatrix} -1 & -12 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} s^i & 0 \\ 0 & s^i \end{bmatrix} + \begin{bmatrix} -1 & -12 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} s^i - 1 & 0 - 12 \\ 0 + 1 & s^i + 6 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = 0$$



$$\begin{bmatrix} \lambda_i - 1 & -12 \\ 1 & \lambda_i + 6 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = 0 \quad \text{--- eq. (i)}$$

Put the value  $\lambda_i = -3$  in eq. (i)

$$\begin{bmatrix} -3 - 1 & -12 \\ 1 & -3 + 6 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -4 & -12 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = 0$$

$$-4c_1 - 12c_2 = 0$$

$$c_1 + 3c_2 = 0$$

Now assume  $c_2 = 1$

$$c_1 + 3c_2 = 0$$

$$c_1 + 3(1) = 0$$

$$c_1 + 3 = 0$$

$$\boxed{c_1 = -3}$$

$$c_1 + 3c_2 = 0$$

$$-3 + 3c_2 = 0$$

$$\boxed{c_2 = 1}$$



Thus, the eigenvector  $C_1$  corresponding to  $\lambda_1 = -3$  is

$$C_1 = \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

First complementary function

$$Y_{1C} = k_1 \begin{bmatrix} -3 \\ 1 \end{bmatrix} e^{-3t} = \begin{bmatrix} -3k_1 e^{-3t} \\ k_1 e^{-3t} \end{bmatrix}$$

Now Next Step:

Put the value of  $\lambda_2 = -2$  in eq. (i)

$$\begin{bmatrix} -2-1 & -12 \\ 1 & -2+6 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -3 & -12 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = 0$$

$$-3C_1 - 12C_2 = 0$$



$$C_1 + 4C_2 = 0$$

Now assume  $C_2 = 1$

$$C_1 + 4C_2 = 0$$

$$C_1 + 4(1) = 0$$

$$C_1 + 4 = 0$$

$$\boxed{C_1 = -4}$$

$$C_1 + 4C_2 = 0$$

$$-4 + 4C_2 = 0$$

$$\boxed{C_2 = 1}$$

Thus, the eigenvector  $C_2$  corresponding

to  $\lambda_2 = -2$  is

$$C_2 = \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = \begin{bmatrix} -4 \\ 1 \end{bmatrix}$$

Second complementary function

$$Y_2 C = K_2 \begin{bmatrix} -4 \\ 1 \end{bmatrix} e^{-2t} = \begin{bmatrix} -4K_2 e^{-2t} \\ K_2 e^{-2t} \end{bmatrix}$$



## Complete Complementary Function

$$y_1(t) = -3k_1 e^{-3t} - 4k_2 e^{-2t}$$

$$y_2(t) = k_1 e^{-3t} + k_2 e^{-2t}$$

## Particular Integral

$$y_p = A^{-1} B$$

where,

$$A = \begin{bmatrix} -1 & -12 \\ 1 & 6 \end{bmatrix}, B = \begin{bmatrix} 60 \\ 36 \end{bmatrix}$$

$$A^{-1} = \frac{\text{Adj } A}{|A|}$$

$$\text{Adj } A = \begin{bmatrix} 6 & 12 \\ -1 & -1 \end{bmatrix}$$

$$|A| = \begin{vmatrix} -1 & -12 \\ 1 & 6 \end{vmatrix}$$



$$= (-1 \times 6) - (1 \times -12)$$

$$= -6 + 12$$

$$= 6$$

$$\boxed{|A| = 6}$$

$$A^{-1} = \frac{1}{6} \begin{bmatrix} 6 & 12 \\ -1 & -1 \end{bmatrix}$$

$$A^{-1} B$$

$$= \frac{1}{6} \begin{bmatrix} 6 & 12 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} -60 \\ 36 \end{bmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} 6 \times -60 + 12 \times 36 \\ -1 \times -60 + -1 \times 36 \end{bmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} 12 \\ 24 \end{bmatrix}$$

$$\therefore p = \begin{bmatrix} 12 \\ 4 \end{bmatrix}$$



## complete general solution

$$y(t) = y_c + y_p$$

$$y_1(t) = -3ke^{-3t} - 4k_2e^{-2t} + 12$$

$$y_2(t) = ke^{-3t} + k_2e^{-2t} + 4$$

Now apply the condition  
on eq. (a) & (b)

$$x_0 = 1.3$$

$$y_0 = 4$$

$$-3k_1 - 4k_2 + 12 = 1.3 \quad \text{--- eq. (i)}$$

$$k_1 + k_2 + 4 = 4 \quad \text{--- eq. (ii)}$$

Divide the eq. (ii) with 4

$$\frac{k_1 + k_2 + 4 = 4}{4}$$

$$k_1 + k_2 = 0 \quad \text{--- eq. (i)}$$

$$4k_1 + 4k_2 = 0 \quad \text{--- eq. (ii)}$$



Solve the equation (ii) & (iv)  
by simultaneous:

$$-3k_1 - 4k_2 = 1$$

$$4k_1 + 4k_2 = 0$$

$$\boxed{k_1 = 2}$$

$$-3k_1 - 4k_2 = 1$$

$$-3(1) - 4k_2 = 1$$

$$-3 - 4k_2 = 1$$

$$-4k_2 = 1 + 3$$

$$-4k_2 = 4$$

$$k_2 = \frac{4}{-4}$$

$$\boxed{k_2 = -1}$$

Putting these values in eq. (a) & (b)

$$y_1(t) = -3k_1 e^{-3t} - 4k_2 e^{-2t} + 12$$

$$y_2(t) = k_1 e^{-3t} + k_2 e^{-2t} + 4$$



$$(b) \quad \begin{aligned} x'(t) - 2x(t) + 3y(t) &= 10 \\ y'(t) - x(t) + 2y(t) &= 9 \end{aligned}$$

$$[ \text{with } x(0)=8, y(0)=5 ]$$

**Solution:**

$$x'(t) - 2x(t) + 3y(t) = 10$$

$$y'(t) - x(t) + 2y(t) = 9$$

Write in matrix equation:

$$A_1 y' + A_2 y = B$$

where the matrices are

$$A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, y' = \begin{bmatrix} x'(t) \\ y'(t) \end{bmatrix}, A_2 = \begin{bmatrix} -2 & 3 \\ -1 & 2 \end{bmatrix},$$

$$y = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}, B = \begin{bmatrix} 10 \\ 9 \end{bmatrix}$$



$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x'(t) \\ y'(t) \end{bmatrix} + \begin{bmatrix} -2 & 3 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} 10 \\ 9 \end{bmatrix}$$

For  $\lambda_1, \lambda_2$  use the formula

$$|A_1 \lambda_i + A_2| = 0$$

$$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \lambda_i + \begin{bmatrix} -2 & 3 \\ -1 & 2 \end{bmatrix} = 0$$

$$\begin{vmatrix} \lambda_i & 0 \\ 0 & \lambda_i \end{vmatrix} + \begin{bmatrix} -2 & 3 \\ -1 & 2 \end{bmatrix} = 0$$

$$\begin{vmatrix} \lambda_i - 2 & 0 + 3 \\ 0 - 1 & \lambda_i + 2 \end{vmatrix} = 0$$

$$\begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$(\lambda - 2)(\lambda + 2) - (-1 \times 3)$$

$$\lambda^2 + 2\lambda - 2\lambda - 4 + 3 = 0$$

$$\lambda^2 - \lambda - 1 = 0$$

Factorization:

$$\lambda(\lambda - 1) + 1(\lambda - 1) = 0$$



$$(x-1) = 0 \Rightarrow \boxed{x_1 = +1}$$

$$(x+1) = 0 \Rightarrow \boxed{x_2 = -1}$$

Complementary Function

For  $C_1, C_2$  we use eigenvector formula:

$$(A - \lambda_i I + A_2) C_i = 0$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \lambda_i + \begin{bmatrix} -2 & 3 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} \lambda_i & 0 \\ 0 & \lambda_i \end{bmatrix} + \begin{bmatrix} -2 & 3 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} \lambda_i - 2 & 0 + 3 \\ 0 - 1 & \lambda_i + 2 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} \lambda_i - 2 & 3 \\ -1 & \lambda_i + 2 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = 0 \quad \text{--- eq. (i)}$$



Putt the value  $x_1 = 1$  in eq. (i)

$$\begin{bmatrix} 1-2 & 3 \\ -1 & 1+2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -1 & 3 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = 0$$

$$-c_1 + 3c_2 = 0$$

$$-c_1 + 3c_2 = 0$$

Now assume  $c_2 = 1$

$$-c_1 + 3c_2 = 0$$

$$-c_1 + 3(1) = 0$$

$$-c_1 + 3 = 0$$

$$+c_1 = +3$$

$$\boxed{c_1 = 3}$$

$$-c_1 + 3c_2 = 0$$

$$-3 + 3c_2 = 0$$

$$\boxed{c_2 = 1}$$



Thus, the eigenvector  $c_1$  corresponding to  $\lambda_1 = 1$  is

$$c_1 = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

First complementary function

$$Y_1 c = k_1 \begin{bmatrix} 3 \\ 1 \end{bmatrix} e^{1t} = \begin{bmatrix} 3k_1 e^t \\ k_1 e^t \end{bmatrix}$$

Next Step:

Put the value  $\lambda_2 = -1$  in eq. (i)

$$\begin{bmatrix} -1-2 & 3 \\ -1 & -1+2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -3 & 3 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = 0$$

$$-3c_1 + 3c_2 = 0$$

$$-c_1 + c_2 = 0$$



Now assume  $C_2 = 1$

$$-3C_1 + 3C_2 = 0$$

$$-3C_1 + 3C_2 = 0$$

$$-3C_1 + 3(1) = 0$$

$$-3C_1 + 3 = 0$$

$$\boxed{C_1 = 1}$$

$$-3C_1 + 3C_2 = 0$$

$$-3(1) + 3C_2 = 0$$

$$-3 + 3C_2 = 0$$

$$\boxed{C_2 = 1}$$

Thus, the eigenvector  $C_2$  corresponding to  $\lambda_2 = \frac{pu}{cp}$  is

$$C_2 = \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Second Complementary function

$$Y_2C = k_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-1t} = \begin{bmatrix} k_2 e^{-1t} \\ k_2 e^{-1t} \end{bmatrix}$$



## Complete Complementary Function

$$y_1(t) = 3ke^t + ke^{-t}$$

$$y_2(t) = ke^t + ke^{-t}$$

## Particular Integral

$$y_p = A^{-1} B$$

where,

$$A = \begin{bmatrix} -2 & 3 \\ -1 & 2 \end{bmatrix}, B = \begin{bmatrix} 10 \\ 9 \end{bmatrix}$$

$$A^{-1} = \frac{\text{Adj } A}{|A|}$$

$$\text{Adj } A = \begin{bmatrix} 2 & -3 \\ 1 & -2 \end{bmatrix}$$

$$|A| = \begin{vmatrix} -2 & 3 \\ -1 & 2 \end{vmatrix}$$



$$\begin{aligned}
 &= (-2 \times 2) - (-1 \times 3) \\
 &= -4 + 3 \\
 &= -1
 \end{aligned}$$

$$|A| = -1$$

$$A^{-1} = \frac{\text{Adj } A}{|A|} = \frac{1}{-1} \begin{bmatrix} 2 & -3 \\ 1 & -2 \end{bmatrix}$$

$$y_p = A^{-1} B$$

$$= \frac{1}{-1} \begin{bmatrix} 2 & -3 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 10 \\ 9 \end{bmatrix}$$

$$= \frac{1}{-1} \begin{bmatrix} (2 \times 10) + (-3 \times 9) \\ (1 \times 10) + (-2 \times 9) \end{bmatrix}$$

$$= \frac{1}{-1} \begin{bmatrix} 20 & -27 \\ 10 & -18 \end{bmatrix}$$

$$= \frac{1}{-1} \begin{bmatrix} -7 \\ -8 \end{bmatrix}$$



$$V_p = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

Complete General Solution

$$Y_t = Y_c + V_p$$

$$Y_1(t) = 3k_1 e^t + k_2 e^{-t} + 7$$

$$Y_2(t) = k_1 e^t + k_2 e^{-t} + 8$$

Now apply the condition on  
eq. (a) & (b)

$$X_0 = 8$$

$$Y_0 = 5$$

$$3k_1 + k_2 + 7 = 8 \quad \text{--- (ii)}$$

$$k_1 + k_2 + 8 = 5 \quad \text{--- (iii)}$$

$$3k_1 + k_2 = 8 - 7$$

$$k_1 + k_2 = 5 - 8$$



$$= 3k_1 + k_2 = 1 \quad \text{--- (iv)}$$

$$k_1 + k_2 = -3 \quad \text{--- (v)}$$

Multiply the eq. (v) with 3

$$k_1 + k_2 = -3$$

$$3k_1 + 3k_2 = -9 \quad \text{--- (vi)}$$

Solve the eq. (iv), (vi) by simultaneous:

$$\begin{array}{r} 3k_1 + k_2 = 1 \\ -3k_1 + 3k_2 = -9 \\ \hline -2k_2 = 10 \end{array}$$

$$k_2 = \frac{10 \times 5}{-2}$$

$$\boxed{k_2 = -5}$$

Put the value  $k_2$  in eq. (iv)

$$3k_1 - 5 = 1$$

$$3k_1 = 1 + 5$$

$$3k_1 = 6$$

$$\boxed{k_1 = 2}$$



Putt the value in eq. (a) & (b)

$$y_1(t) = 3(2)e^t + (-5)e^{-t} + 7$$

$$y_2(t) = 2e^t + (-5)e^{-t} + 8$$

$$y_1(t) = 6e^t - 5e^{-t} + 7$$

$$y_2(t) = 2e^t - 5e^{-t} + 8$$

